



Our Website

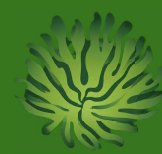
Demo Video



TEAM 2: ALGAE FIGHTERS

Peter Quawas, Daniel Chen, Anton John Del Mar

ECE 196 - Fall 2024



UC San Diego

JACOBS SCHOOL OF ENGINEERING

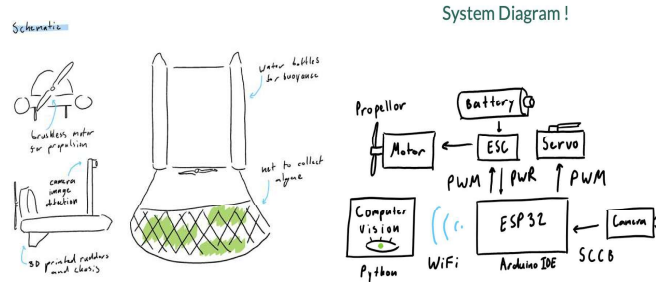
Problem Definition

The UCSB Lagoon is often covered in toxic algae which makes the water unsafe for humans and marine life.



Proposed Idea

A small floating boat that will navigate the water detect algae using computer vision while dragging a net behind it to collect the algae.



Hypothesis

The boat will:

- Float
- Self navigate through water
- Detect algae blooms using Image recognition
- Collect algae using a mounted net
- Be stable on the water

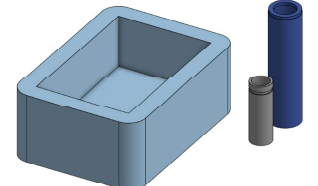
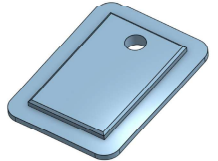
Prototype

Our prototype boat was too thick and heavy and the lid was too loose. We fixed this in our final design.



Printed boat with Lid and propeller mount

Initial Lid design with hole for camera stand



Initial boat and camera stands

Test Results



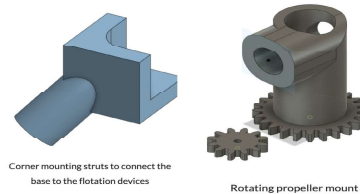
Propellor and Servo test



First Water test

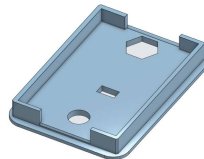
Final Design

The boat's entire body will be 3D printed along with its struts which will use PVC pipes to connect to the flotation devices.

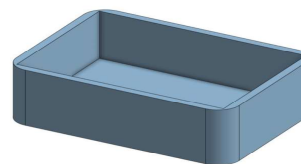


Corner mounting struts to connect the base to the flotation devices

Rotating propeller mount

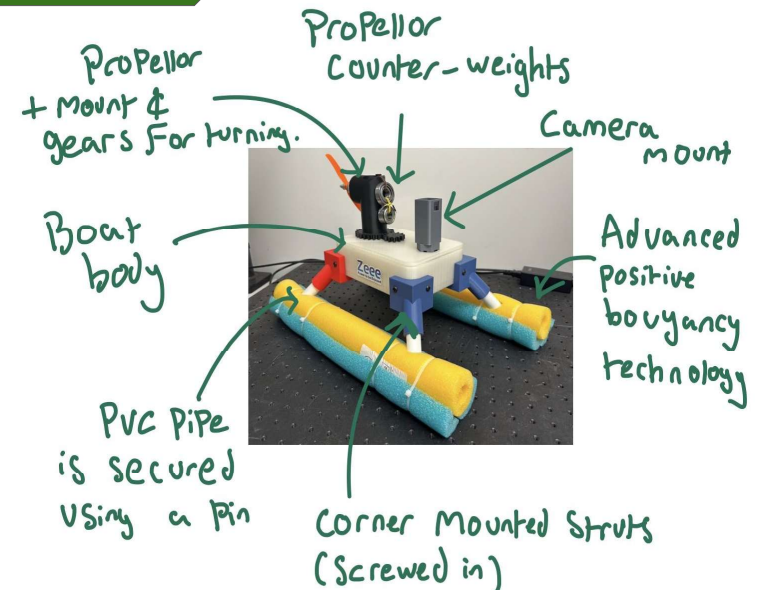


Boat Lid with holes for the Camera mount, small gear servo, and propeller stand



Boat body to store components safely

Our Solution



Milestones

OCTOBER

25th

Define the problem and brainstorm solutions

28th

Working Algae detection Algorithm

1st

Create a boat frame that can float

4th

The boat can make it straight across a pool

11th

Install a steering mechanism that can turn the boat 180°

NOVEMBER

18th

ESP can communicate wirelessly and turn on an LED

20th

Custom PCB allows camera and ESP to use the Algae detection code and turn the boat

29th

Successfully navigate a pool and pick up mock algae using a net